SOP MANUFACTURING AND ASSEMBLY (IN-HOUSE)

[CONFIDENTIAL]
[UTTHUNGA]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	0.1	Initial draft	New	Santosh V 25-Apr-25		
2	0.2	Review			Sanmangala 30-Apr-25	
3	Issue B V1.0	Approval of document for IMS release				Sanmangala 30-Apr-25
4	Issue BV.1.1	Section 9.4 e added Included Section 12	Included Safe keeping of critical information	Santosh V 07-Jul-25	Anitha N 07-Jul-25	Sanmangala 11-Jul-25

1 Purpose

This SOP defines the production management process when manufacturing is done in-House. The production adheres to approved design specifications. The production process meets established quality standards and deadlines. The product quality aligns with the Quality Control (QC) documentation.

2 Scope

This SOP applies to in-House production activities for products designed and developed by the company as well as turnkey projects solely undertaken for production.

For Outsources production, refer SOP Production process UT/SD/EP/SOP/01

3 Abbreviations

a. PDS: Product Design Specifications

b. BOM: Bill of Materials

c. CAD: Computer Aided Design

d. D&D: Design and Development

e. ECR: Engineering Change Request

f. ECO: Engineering Change Order

g. CAPA: Corrective and Preventive Action

h. QC: Quality Control

i. QMS: Quality Management System

j. EMS: Electronics Manufacturing Services

k. ISO: International Organization for Standardization

l. MSDS: Material Safety Data Sheets

m. NDAs: Non-Disclosure Agreements

4 Terms and Definitions

a. Production plan: Production Planning provides a stepwise solution to pull items to be produced, fetch raw materials through their BOMs and check their availability in warehouses. It enables Production Supervisor to request raw-materials and initiate Work Orders accordingly

b. Work Order: This serves as the central artefact, triggering raw material transfers, generating Job Cards, and enabling backflushing of finished goods. Sales teams can also refer to Work Orders to track production progress and keep customers updated.

c. Prototype: An initial model or sample of a product used to evaluate and validate design specifications.

d. Risk Register: A tool used to document and monitor potential risks and their mitigation strategies.

e. Corrective and Preventive Action (CAPA): A process to identify, address, and prevent the recurrence of quality issues.

f. Engineering Change Request (ECR): A formal request to modify a product's design.

g. Engineering Change Order (ECO): A document that authorizes and details the implementation of a design change.

h. Quality Control (QC): Processes and procedures to ensure products meet specified quality standards.

i. Bill of Materials (BOM): A comprehensive list of materials, components, and assemblies required to build a product.

5 Responsibilities

a. Procurement Team: The Procurement Team is responsible for identifying potential vendors, facilitating audits, and negotiating contracts while ensuring compliance with confidentiality agreements. They manage timely delivery of materials, verify receipt and quality, and maintain procurement documentation to ensure consistency and accuracy.

b. Vendor Assessment Team: This team conducts audits to assess vendors' financial stability, certifications, and production capacity, reviews submitted samples and makes approval or rejection decisions based on audit results to ensure that vendors meet quality and regulatory standards.

c. Design Team: The Design Team provides detailed Product Design Specifications (PDS) and Bill of Materials (BOM), addresses vendor queries about production feasibility, participates in design reviews and validation, documents change through Engineering Change Requests (ECRs), and updates relevant documentation to reflect any revisions.

d. Production Manager/Production Supervisor: The Production Manager/Production Supervisor coordinates design handoff meetings, inspects prototypes, ensures vendors have comprehensive production plans, establishes communication channels for updates, oversees quality control processes, and verifies compliance in packaging and shipment procedures.

e. Project Manager: The Project Manager manages design changes and coordinates with vendors to adjust timelines, oversees quality control through CAPA processes, ensures continuous improvement, and establishes acceptance criteria for final product testing before deployment while collecting feedback for future enhancements.

f. Legal: The legal advisor comes up with terms and conditions, recommends on regulatory and compliance requirements

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Vendor Selection & Approval	Procurement Team, Vendor Assessment Team	Procurement Team	Production Team, Design Team, Legal	Stakeholders, Project Team
Release for Production	Production Manager/Production Supervisor	Production Manager/Production Supervisor, Design Team,	Hardware lead, Vendor	Procurement Team, Project Manager
Risk Management	Project Team, Production Team	Project Manager	Vendors, Procurement Team	Design Team, Senior Management
Production Planning	Production Manager/Production Supervisor	Project Manager	Procurement Team, Vendor	Design Team, Project Team
Raw Material processing	Production Manager/Production Supervisor	Production Manager/Production Supervisor	Design Team, Quality Control team	Project Team, Quality Assurance Team
Component Manufacturing	Production Manager/Production Supervisor	Production Manager/Production Supervisor	Design Team, Quality Control team	Project Team
Sub-Assembly	Production Manager/Production Supervisor	Production Manager/Production Supervisor	Design Team, Quality Control team	Project Team
Final Assemble & Integration	Production Manager/Production Supervisor	Production Manager/Production Supervisor	Design Team, Quality Control team	Project Team, Quality Assurance Team

Handling Changes and Revisions	Production Manager/Production Supervisor	Project Manager	Production Team	All stakeholders involved in the change
Quality Control, Inspection & End of line testing	Quality Control	Production Manager/Production Supervisor	Design Team	Design Team, Quality Assurance
Packaging, Logistics & Shipping	Production Team, Vendor	Production Manager/Production Supervisor	Logistics Provider, Vendor	Procurement Team, Client

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Vendor Assessment Team	Potential vendors, audit checklists	Vendor Selection & Approval	Vendor audit, assessment of samples, Quality control checks, QMS certificates	Approved/ rejected vendor list, signed contracts	Production Team, Project Team
Production Manager /Production Supervisor	Technical documentation, Manufacturing instructions, QC Guidelines, Supply Chain & Logistics	Release for production	Design handoff meeting,	Acceptance by Vendor	Vendor

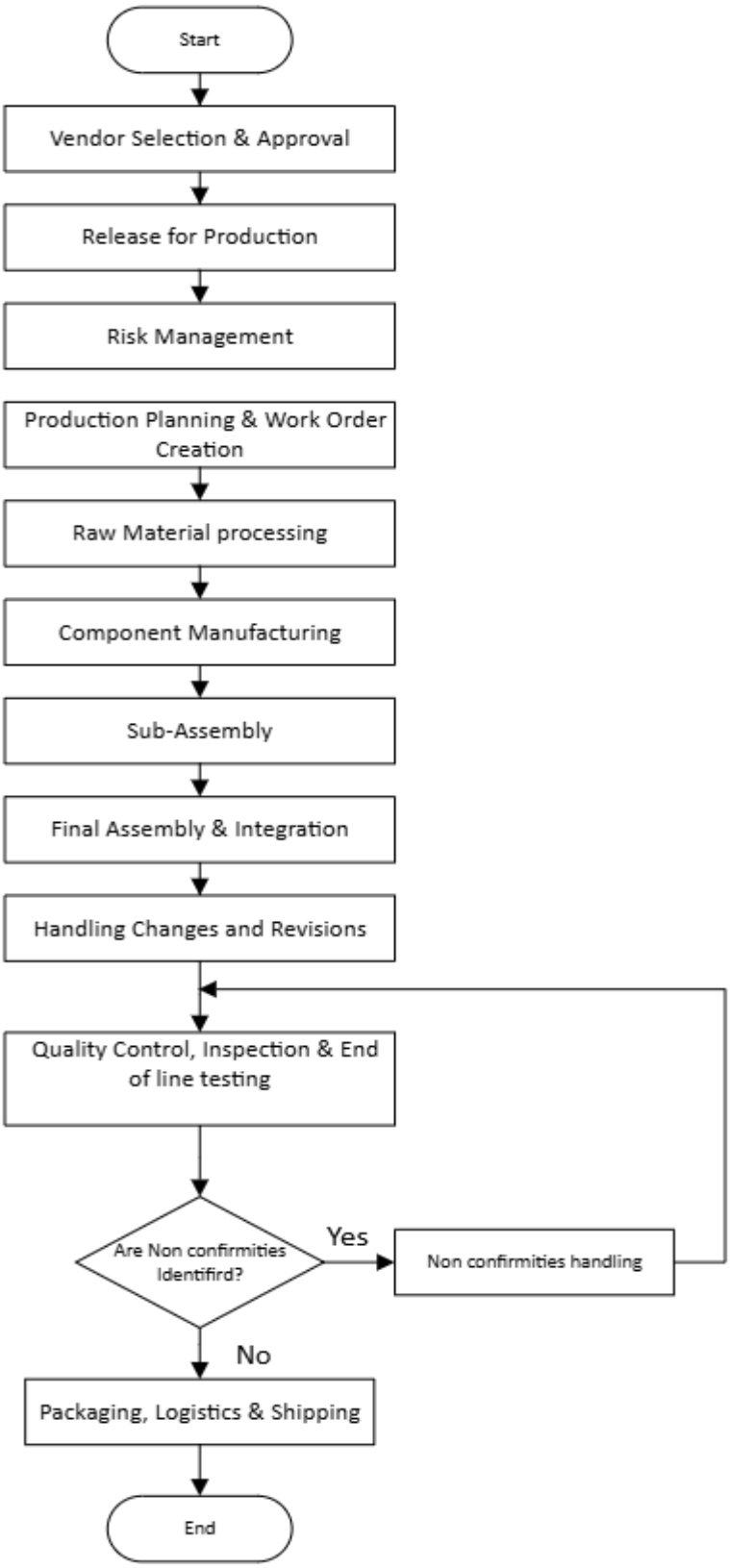
Project Manager	Identified risks, historical data, risk register	Risk Management	Risk assessment meetings, mitigation strategies, monitoring risk register	Updated risk register, mitigation plans	Project Team, Production Team, Procurement Team
Production Manager /Production Supervisor	Production Schedule & Supply Chain Plan	Production Planning	Production plan and milestones, timely material delivery	Production schedule, Material requirement plan, resource allocation plan	Production team, QC team
Vendor, Design team, Production Manager /Production Supervisor	BOM, Engineering drawings, MSDS, Standard Operations	Raw Material processing	In-process inspections, CAPA Processes, regular quality data reviews	Inspection Report, Non-Conformance report	Production team, QC team
Vendor, Design team, Production Manager /Production Supervisor	BOM, Engineering drawings, MSDS, Standard Operations, Inspection and Testing Guidelines	Component Manufacturing	In-process inspections, CAPA Processes, regular quality data reviews	Quality Inspection Report, Rework/Non-Conformance Report	QC team
Vendor, Design team, Production Manager /Production	BOM, Engineering drawings, MSDS, Standard Operations, Inspection and Testing Criteria,	Sub-Assembly	In-process inspections, CAPA Processes, regular quality data reviews	Quality Inspection Report, Rework/Non-Conformance Report	, QC team

Supervisor	Electrical & Wiring Schematics				
Vendor, Design team, Production Manager /Production Supervisor	BOM, Engineering Drawings & Assembly Instructions, Standard Operations, Electrical & Wiring Schematics, Firmware & Software Configuration Guidelines, Testing & Inspection Criteria	Final Assembly & Integration	Inspection, validation, CAPA Processes, regular quality data reviews	Finished product, Quality Inspection & Functional Validation, Firmware/Software Installation Log, Rework/Non-Conformance Report	QC team
Project Manager	ECR, ECO, Updated design documents, BOM, Cost estimate/Change impact Analysis	Handling Changes and Revisions	Approved changes to process, cost, Vendor Notification	Approved ECR/ECO, Updated BOMs and Assembly Instruction, Revised Production Schedules, Cost, Approval Records	Production Manager/Production Supervisor, Vendors
Production team, Quality Assurance team	Work Instructions & Process Control Plans, Test Plans & Acceptance Criteria	Quality Control, Inspection & End of line testing	In-Process inspection, Final inspection, Testing, CAPA process	In-Process Inspection Report, Final Quality Inspection Report, Non-Conformance Report, Regulatory Compliance	Production team

				& Certification Records	
Production Team, Design team	Finished products, packaging standards, Final Inspection Report, Labeling Work Instructions, Regulatory & Customer-Specific Labeling Requirements, Serial Number & Barcode Allocation List	Packaging, Logistics & Shipment	Packaging according to standards and tracking	Packaged and labeled products, shipment tracking info, Regulatory Compliance Labeling Record	Customers, Logistics Providers, Project manager

8 Overall Process Flow Chart

a. Manufacturing and Assembly



9 Process Description

Each manufacturing and assembly process must be executed in accordance with production requirements, ensuring efficiency, compliance, and quality assurance. The applicability of specific subprocesses outlined in this SOP shall be determined based on the sourcing strategy and the stage at which production activities commence.

Where components are procured from **approved vendors** and meet the required **quality and compliance standards**, production shall commence directly from the **Sub-Assembly** stage, rendering the preceding steps non-mandatory. Similarly, if pre-assembled units are utilized for final product assembly, the process shall begin from the **Final Assembly & Integration** stage, bypassing prior sub-processes.

It is the responsibility of the **Production Supervisor** to assess production inputs and determine the applicable starting point in alignment with **Bill of Materials (BOM) using** template UTSFPURTMP10 & UTSFPURTMP13**, engineering specifications, and procurement plans**. The **Quality Control Team** shall verify that all incoming materials conform to the established **inspection criteria** before integrating them into the designated stage of manufacturing.

Each stage must be executed as per the Standard Operating Procedures for the Product and the Standard Operations defined for each step of product manufacturing, ensuring adherence to process controls, quality checkpoints, and compliance standards.

9.1 Supply Chain Management

a. Procurement team identifies vendors as per Procurement SOP UTFPROSOP for Raw Materials and Components
   - Review the product design and determine the raw materials and components required.
   - Create a list of specifications, including part numbers, quantities, lead times, and quality standards.
   - Prioritize critical materials/components based on product functionality and production schedules.
   - Research and Shortlist Potential Vendors
     - Conduct market research to identify potential suppliers

b. Evaluate vendor reputation, financial stability, delivery performance, and certifications based on supplier evaluation process

c. Use Vendor evaluation form UTSFPURTMP05

9.2 Release for production

a. Design team provides the Production Manager/Production Supervisor the following technical documentation for production

Category	Required Documents & Information
Technical Documentation	BOM, CAD Drawings, Firmware & software files, Schematics & Circuit Diagrams
Manufacturing Instructions	Assembly Instructions, Testing Procedures, Packaging Details
Quality & Compliance	QC Standards, Compliance & Certifications, Tolerance & Acceptance Criteria
Supply Chain & Logistics	Supply Chain & Logistics Information, Procurement & Lead Time Information, Shipping & Logistics Coordination details

- Bill of Materials (BOM)
  - List of all components, subassemblies, and raw materials. Use Bill of Material template UTSFPURTMP10 & UTSFPURTMP13
  - Unit cost estimates
  - Component tolerances & material specifications
- CAD Drawings
  - Mechanical Drawings
  - PCB Layout Files (Gerber files for circuit boards)
  - Exploded Views (Assembly structure & component arrangement)
  - Dimensional & Tolerance Specifications
- Firmware & Software Files
  - Embedded system code & firmware (C, Python, or compiled HEX/BIN files)
  - Microcontroller bootloader & communication protocols
  - API documentation for connectivity
  - Use for release for production UTSFPURTMP28
  - Manufacturing Instructions
  - Ensure the security measures are in place to ensure that files are secure during transmission and post receive by vendor. Implement appropriate measures like Digital signatures, secure files transfer, access control, validation of receipt, secure hardware elements, Cryptographic Hash Validation, Secure Boot / Trusted Platform Module, legal controls, Encrypted Formats as appropriate.
- Schematics & Circuit Diagrams
  - Electronic design schematics

    - Power distribution, signal paths, and component placement
- Assembly Instructions
    - Step-by-step process for putting together components
    - Sequence of assembly (mechanical first, then electronic, etc.)
    - Soldering, welding, or bonding instructions
- Testing Procedures
    - Functional testing scripts (automated & manual)
    - Pass/fail criteria for each stage
    - Environmental & stress testing requirements
- Packaging Guidelines
    - Final product packaging (box design, inserts, padding materials)
    - Retail labeling, barcodes, and QR codes
    - Shipment packaging for bulk orders

b. Quality & Compliance Requirements

- Quality Control (QC) Standards
    - Incoming material inspection procedures Refer SOP – Incoming Material Inspection UT/PS/IGI/SOP
    - In-process & final product quality checks
    - Sampling plan & rejection criteria (AQL standards)
- Compliance & Certifications
    - Industry standards (CE, FCC, ATEX, RoHS, ISO 9001, UL, etc.)
    - Product safety requirements (shock, fire, electrical hazard compliance)
    - Country-specific regulations (FDA for medical devices, BIS for India, etc.)
- Tolerance & Acceptance Criteria
    - Mechanical tolerances
    - Electronic tolerances (Voltage, current limits)
    - Cosmetic defect tolerances (scratches, color variations)
- Supply Chain & Logistics Information
    - Approved Supplier List (ASL)
    - List of preferred suppliers for critical components
    - Alternatives in case of shortages
- Procurement & Lead Time Information
    - Component availability & expected delivery times
    - Minimum order quantities (MOQs) for bulk purchases
- Shipping & Logistics Coordination details

- Incoterms (FOB, CIF, DDP)
- Storage conditions (humidity, temperature limits for electronics)
- Labeling & tracking requirements

9.3 Risk Management

a. The Production Manager/Production Supervisor with project manager with inputs from vendor and procurement does risk assessment as per SOP - Risk & Opportunity Management UT/IM/RMP/SOP and comes up with mitigation and contingency plan as applicable.

b. Takes action as per the mitigation and contingency plan.

9.4 Production Planning & Work Order Creation

a. Production planning is the **initial stage** of the manufacturing process, ensuring that all production activities are strategically scheduled, resourced, and aligned with delivery commitments, quality standards, and operational efficiency. It involves demand forecasting, resource allocation, sequencing of manufacturing activities, and generation of work orders.

The following must be available to begin planning:

- **Customer Order / Sales Order** – Defines production demand and delivery schedule.
- **Bill of Materials (BOM)** – Lists required materials, components, and procurement details.
- **Engineering Drawings & Product Specifications** – Provides design and technical requirements.
- **Inventory Report** – Verifies stock availability for production.
- **Manufacturing Capacity Report** – Assesses machine and workforce availability.

b. Production planning must be completed before **any subsequent manufacturing step**, ensuring that production resources, materials, and schedules are **optimized** to meet project timelines and quality requirements.

Key Considerations for Production Planning

- **Bill of Materials (BOM) Review** – Identify all required components, materials, and procurement sources.
- **Engineering Drawings & Specifications** – Define product structure, design requirements, and quality parameters.
- **Inventory & Procurement Status** – Verify availability of raw materials, components, and sub-assemblies.

- **Capacity Planning** – Assess machine availability, labor allocation, and production schedules.
- **Regulatory & Compliance Requirements** – Ensure adherence to industry and organizational compliance standards.
- **Production Priority & Lead Time** – Align production start dates with customer deadlines and project priorities.

c. Production Manager/Production Supervisor creates the Production Schedule & Supply Chain Plan and shares with the Vendor. As part of planning does the following activities.
   - Set Production Milestones: Create a timeline with clear milestones for prototyping, pilot runs, full-scale production, and product testing.
   - Define Lead Times: Consider lead times for raw material procurement, component delivery, and manufacturing time

d. Artefacts to be created at this step include:
   - **Production Plan** – Outlines the detailed schedule, resource allocation, and execution sequence. Refer to Annexture Production Plan for the contents of Production Plan.
   - **Work Order** – Authorizes specific production tasks and provides live status of production progress. Refer to Annexture Work Order for contents of Work Order.
   - **Material Requirement Plan (MRP)** – Identifies procurement needs and tracks material availability.
   - **Standard Operations** – Specifies step-by-step procedures for manufacturing the product. These must be product-specific, and if an existing **Standard Operation** is defined, it must be reused. The **Production Supervisor** must ensure that the appropriate **Standard Operations** are available and suitable for the item under production.

e. Critical information in form of hard copies or soft copies in storage devices are stored in fireproof, flood proof, safe. This may include not limited to the following Critical documentation like design information, HR records, financial information, crucial IP, sensitive inventory like prototypes, rare components, regulatory and legal information. This is to mitigate risks from fire or flood and faster recovery of sensitive information, restricts access to relevant personnel

9.5 Raw Material Processing

a. The following inputs must be available before starting raw material processing:

   - **Bill of Materials (BOM)** – Lists all materials required for production.

- **Engineering Drawings** – Specifies dimensions, tolerances, and material requirements.
- **Material Safety Data Sheets (MSDS)** – Provides handling and safety guidelines.
- **Standard Operations** – Details step-by-step procedures for each material type.

Raw material processing is the first stage of manufacturing and involves preparing raw materials to meet design and production specifications. This step ensures that the materials used in the production process are of the required quality, shape, and consistency.

The process includes methods such as cutting, casting, molding, and heat treatment, depending on the material properties and end-use requirements Refer respective guidelines.

Raw material processing must be done before the component manufacturing stage to ensure that all raw materials meet the design specifications outlined in the engineering drawings and BOM.

b. Determine the appropriate processing method based on:
- Material type
- Dimensional requirements
- Mechanical properties
- Final application

c. Evaluate the following factors before selecting the processing method.
- Hardness
- Thermal resistance
- Machinability
- Structural integrity

The Production Supervisor is responsible for overseeing raw material processing, while Production Operators execute the tasks based on defined work instructions. Quality Control inspects the processed materials to ensure compliance with specifications.

d. Artefacts to be created at this step include:

- **Inspection Report** – Logs quality checks conducted on processed materials.
- **Non-Conformance Report (if applicable)** – Documents any deviations from specifications.

9.6 Component Manufacturing

a. The following input documents must be available before starting component manufacturing:

- **Bill of Materials (BOM)** – Lists components and their specifications.
- **Engineering Drawings** – Provide detailed dimensions, tolerances, and finishing requirements.
- **Standard Operations** – Define machining, welding, or finishing techniques for each component.
- **Material Safety Data Sheets (MSDS)** – Outline handling and safety precautions.
- **Inspection and Testing Guidelines** – Specify quality control measures and acceptable tolerances.

b. The following steps needs to be followed for component manufacturing.

- Review Inputs: Verify availability of processed raw materials, Bill of Materials (BOM), engineering drawings, and Standard Operating Procedures (SOPs).
- Select Manufacturing Method: Determine the appropriate manufacturing method—such as machining, welding, soldering, or surface treatment—based on material properties, dimensional tolerances, and functional requirements.
- Set Up Workstations: Prepare machines, tools, and materials required for the selected manufacturing process in accordance with work instructions.
- Execute Component Manufacturing: Perform machining, welding, soldering, or surface finishing operations as specified in engineering drawings and standard work procedures.
- Monitor Process Parameters: Continuously monitor key process parameters such as speed, temperature, pressure, and alignment to maintain quality and efficiency.
- Perform In-Process Inspections: Conduct interim quality checks during the manufacturing process to verify compliance with tolerances and design specifications.
- Log Process Data: Record details of the process execution, including techniques used, measurements taken, and any anomalies encountered.
- Conduct Final Component Inspection: Quality Control inspects each finished component for dimensional accuracy, surface finish, and mechanical integrity against the specified criteria.
- Address Non-Conformances: Document and segregate any non-conforming components. Initiate corrective actions or rework procedures as needed.

c. Approve for Sub-Assembly: Approve conforming components for release to the sub-assembly stage. Ensure traceability and documentation are complete.

- **Quality Inspection Report** – Logs component measurements, tolerances, and defects (if any).
- **Rework/Non-Conformance Report (if applicable)** – Documents deviations from specifications and required corrective actions.

9.7 Sub-Assembly

a. The following input documents must be available before starting sub-assembly:

- **Bill of Materials (BOM)** – Lists all components required for each sub-assembly.
- **Engineering Drawings & Assembly Diagrams** – Provide step-by-step instructions for assembling components.
- **Standard Operations** – Define assembly methods, fastening techniques, and testing procedures.
- **Electrical & Wiring Schematics (if applicable)** – Specify wiring connections, signal paths, and power requirements.
- **Testing & Inspection Criteria** – Outline quality checkpoints, functional validation steps, and measurement tolerances.

b. The following steps needs to be followed for Sub-Assembly.

- Confirm Readiness for Sub-Assembly: Production Supervisor verifies that all components required for sub-assembly have passed inspection and meet specifications outlined in the Bill of Materials (BOM) and engineering drawings.
- Review Assembly Documentation: Examine engineering drawings, assembly diagrams, wiring schematics (if applicable), and standard operating procedures to determine the correct assembly sequence and techniques.
- Prepare Assembly Workstation: Operators sets up tools, jigs, and fixtures required for mechanical fitting, electrical wiring, or electronic integration based on the sub-assembly type.
- Execute Sub-Assembly Tasks: Perform the assembly of mechanical, electrical, or electronic modules by fitting, fastening, soldering, or wiring components according to defined work instructions.
- Perform In-Process Validation: Quality Control conducts functional and visual checks at predefined quality checkpoints during the sub-assembly process to ensure alignment, fit, and proper integration.

- Log Assembly Progress: Operators record assembly steps completed, observations made, and any issues encountered during the sub-assembly process.
- Conduct Final Sub-Assembly Inspection: Quality Control inspects each completed sub-assembly for conformance with dimensional tolerances, functional criteria, and electrical continuity (if applicable).
- Address Defects or Deviations.
- Document any non-conforming assemblies, initiate corrective actions, and rework as required to meet quality standards.
- Approve for Final Assembly: Production Supervisor approves, and releases validated sub-assemblies for transition to the Final Assembly & Integration stage, ensuring all documentation is complete and traceable.

c. Artefacts to be created at this step include:
   - **Quality Inspection Report** – Logs sub-assembly measurements, functionality checks, and any defects.
   - **Rework/Non-Conformance Report (if applicable)** – Documents deviations from specifications and necessary corrective actions.

9.8 Final Assembly & Integration

a. The following input documents must be available before starting final assembly:
   - **Bill of Materials (BOM)** – Lists all required components and sub-assemblies.
   - **Engineering Drawings & Assembly Instructions** – Provide step-by-step guidelines for assembling and integrating components.
   - **Standard Operations** – Define the sequence of operations, fastening methods, and alignment procedures.
   - **Electrical & Wiring Schematics (if applicable)** – Specify wire routing, connector configurations, and power distribution.
   - **Firmware & Software Configuration Guidelines (if applicable)** – Detail flashing procedures, parameter settings, and software dependencies.
   - **Testing & Inspection Criteria** – Outline final validation steps, measurement tolerances, and quality acceptance levels.

b. The following steps needs to be followed for Final assembly and Integration.
   - Begin final assembly only after all sub-assemblies are completed and pass quality inspection.
   - Combine sub-assemblies, components, and system elements to build the complete product.

- Ensure all parts fit correctly, function properly, and meet design specifications before final quality inspection.
- Perform the following tasks during final assembly and integration:
  - Mechanical assembly
  - Electrical integration
  - Firmware installation (if applicable)
  - System validation
- Determine assembly and integration processes based on:
  - Product design
  - Functional requirements
  - Interconnection standards
  - Fastener selection
  - Wiring specifications
  - Firmware/software dependencies
  - Quality acceptance criteria

c. Understand and follow critical requirements as per inputs from design to avoid errors:
   - Assembly sequence
   - Mechanical tolerances
   - Electrical continuity checks
   - System configuration

   The Production Supervisor is responsible for overseeing final assembly and integration, ensuring all procedures are followed, and verifying alignment with product specifications. Production Operators execute the assembly, wiring, and integration tasks. Quality Control performs final inspections to confirm compliance with product requirements. Software Engineers (if applicable) handle firmware installation, system programming, and functional validation.

d. Artefacts to be created at this step include:
   - **Quality Inspection & Functional Validation Report** – Logs final product testing, performance checks, and compliance verification.
   - **Firmware/Software Installation Log (if applicable)** – Documents installed versions, configurations, and validation results.
   - **Rework/Non-Conformance Report (if applicable)** – Captures deviations from specifications and required corrective actions.

9.9 Handling Changes and Revisions

a. The following input documents must be available for handling changes

- **Engineering Change Request (ECR) Form**– Ensures all components and materials match specifications.
- **Engineering Change Order (ECO)** – Formal document authorizing the change after ECR approval.
- **Updated Design Documents**– Revised drawings, CAD models, specifications
- **Original Bill of Materials (BOM)**– For comparison and modification.
- **Cost Estimates/Change Impact Analysis**– Documents detailing potential cost or operational impacts

b. The following steps needs to be followed for handling changes and revisions

- Project Manager communicates all design changes through Engineering Change Requests (ECRs) ENGINEERING CHANGE REQUEST FORM - ECR or Engineering Change Orders (ECOs) follow SOP – Engineering Change Request Management UT/SD/ECRM/SOP
- Informs the Production Manager/Production Supervisor of any design changes and adjusts production timelines, as necessary.
- Production Manager/Production Supervisor assess the scope of design changes to determine if the production needs to submit a new prototype or perform validation runs before resuming production.
- Ensure all design changes are clearly documented and approved before implementation.
- Production Manager/Production Supervisor communicate any potential impact on handling charges due to design changes to Project Manager
- Monitor and track all revisions to ensure they are accurately reflected in production processes.
- Production Manager/Production Supervisor updates any affected documentation, such as bill of materials (BOM) and assembly instructions.
- Review and approve any additional costs associated with design changes and revisions.
- Establish a process for resolving any discrepancies or issues that arise from design changes.

c. Artefacts to be created at this step include:

- **Approved ECR/ECO**– Signed-off document confirming the change.

- **Updated BOMs and Assembly Instruction**– Reflecting the approved design modifications
- **Revised Production Schedules** – Adjusted to accommodate changes
- **Change Impact Report**– Details on handling charges, timelines, or resource adjustments.
- **Cost Approval Records -** Signed documents confirming approval of additional costs

9.10 Quality Control, Inspection & End of line testing

a. The following input documents must be available before starting quality control and inspection:

- **Bill of Materials (BOM)** – Ensures all components and materials match specifications.
- **Engineering Drawings & Product Specifications** – Define dimensional tolerances, assembly requirements, and acceptable quality limits.
- **Work Instructions & Process Control Plans** – Outline the standard operating procedures for quality checks at each stage.
- **Regulatory & Certification Standards** – Specify industry requirements that the product must meet.
- **Test Plans & Acceptance Criteria** – Detail functional, mechanical, and environmental test parameters.

b. The following steps needs to be followed for Quality Control, Inspection & End of line testing

- In-Process Quality Control: Quality control and inspection ensure that the manufactured and assembled product meets design, functional, and safety requirements before release. This process involves verifying dimensional accuracy, material integrity, functional performance, and compliance with regulatory standards. It includes in-process inspections, final product validation, and regulatory compliance checks.
- Quality control and inspection must be performed at multiple stages throughout manufacturing, including raw material processing, component manufacturing, sub-assembly, final assembly, and before packaging. The inspection type is determined based on the product's specifications, regulatory requirements, customer expectations, and internal quality standards. Key factors to consider include dimensional tolerances, surface finish, material properties, functional validation, electrical continuity, firmware compatibility, and environmental conditions where the product will be used.
- The Quality Control (QC) team is responsible for executing inspections and functional testing. The Quality Assurance (QA) Manager oversees the inspection process and ensures compliance with standards. Production

Operators assist by performing self-checks and addressing non-conformances SOP - Control of Non-Conforming Product UT/PS/CNP/SOP. Regulatory Compliance Officers (if applicable) ensure that the product meets industry standards like ISO, CE, RoHS, and UL.

- End-of-Line Testing:
  - Quality Control team performs comprehensive tests on the finished product to verify functionality (e.g., electronic testing, pressure tests for seals).
  - Ensure compliance with relevant standards and specification provided by Design team

c. Artefacts to be created at this step include:

- **In-Process Inspection Report** – Records quality checks conducted during manufacturing and assembly.
- **Final Quality Inspection Report** – Logs final product testing, compliance verification, and test results.
- **Non-Conformance Report (if applicable)** – Documents deviations from specifications and required corrective actions.
- **Regulatory Compliance & Certification Records (if applicable)** – Maintains documentation for industry certification and audit purposes.

9.11 Product Packaging, Labeling & Shipping

a. The following input documents must be available before starting the labeling process:

- **Final Inspection Report** – Confirms that the product meets quality and regulatory standards.
- **Labeling Work Instructions** – Define label type, placement, and required information.
- **Regulatory & Customer-Specific Labeling Requirements** – Ensure compliance with standards such as CE, RoHS, UL, ISO, or other applicable certifications.
- **Serial Number & Barcode Allocation List** – Ensures accurate tracking and traceability.

b. The following steps needs to be followed for product packaging, Labelling & Shipping

- Perform product identification and certification labeling immediately after final quality inspection and before storage or shipment.
- Apply identification labels, serial numbers, barcodes, and certification marks as per Packing & Shipment SOP UT/PS/PAS/SOP.
- Ensure labeling supports traceability, regulatory compliance, and warranty validation.

- Determine labeling requirements based on:
  - Product specifications
  - Industry regulations
  - Customer-specific requirements
  - Production Team Responsibilities:

Affix all required identification and certification labels to the product.
The Production Team is responsible for affixing identification and certification labels, while the Quality Control (QC) Team verifies accuracy and compliance including all certification markings meet legal and industry standards.

c. Artefacts to be created at this step include:
   - **Regulatory Compliance Labeling Record** – Documents certification markings and regulatory approvals.
   - **Non-Conformance Report (if applicable)** – Identifies incorrect or missing labels requiring correction.

For details on **product packaging before shipment**, refer to the **Packaging and Shipment SOP**.

10 Information Security requirements

a. 10.1 Suppliers shall be assessed based on adherence to information security requirements specified by the Utthunga.

b. Supplier shall sign the NDA with the organization if required. NDA is applicable for all service-based suppliers, it is also applicable when confidential data such as specifications, drawings, designs, patterns are to be shared to the suppliers (subcontractors, consultants & testing)

c. ICT (Information Communications Technology) based suppliers shall also be monitored on agreed SLA's (Service Level Agreements

d. Disaster Recovery and Business Continuity planning shall be in place.

11 References

e. ISO 9001 Clauses 8.1

f. ISO 27001 Clauses 8.1, Controls A.14

g. ISMS Control Policies

h. SOP – Procurement

i. SOP – production

j. SOP- Packaging and Shipment

12 Metrics

a. On-Time Delivery Rate

b. First Pass Yield

c. Capacity Utilization

d. Number non-conforming products

e. Number of customer complaints

f. Number of audit findings

g. % of scheduled maintenance activities completed on time.

h. Number of safety incidents/near misses

i. Injuries leading to lost time per thousand hours worked.

13 Annexures

a. Production Plan

Production Plan shall have unique ID in the form MFG-PP-YYYY-XXXXXX (where YYYY is year and XXXXXX is sequence number starting from 000001) it will contain the below details. Applicable section of Production plan to be updated based on the scope by the Production Supervisor.

- Posting Date:
- Assembly items:
  - No.
  - Item Code
  - BOM No.
  - Planned items
  - UOM
  - Finished Goods
  - Planned Start Date (mm-dd-yyyy hh:mm)
- Sub Assembly Items
  - No.
  - Sub assembly item code
  - Scheduled time (mm-dd-yyyy hh:mm)
  - Required Quantity

- BOM No.
- Manufacturing Type (In house/Outsourced)

- Row Material Warehouse
- Row Materials
  - No.
  - Item Code
  - Plan to request
  - Quantity as per BOM
  - Quantity in Stock

b. Work Order

Work Order Shall contain the following details.

- Work Order will have unique id in the form: MFG-WO-YYYY-XXXXXX (where YYYY is year and XXXXXX is sequence number starting from 000001)
- Status: <New, In progress, Completed>
- Item to Manufacture: <Description of the item>
- Item Name: <Name of product manufactured>
- BOM No:<Corresponding BOM Number>
- Sales Order:<Corresponding Sale Order Number>
- Quantity To Manufacture: <Number>
- Manufactured Quantity:<Number>
- Standard Operations details: <Standard operations for each step >

14 Forms

a. SOS Template

b. Work Order Template

c. Production Plan Template